This document describes an efficient method to marginalize likelihood-ratio $\mathcal{L}$ over luminosity distance $D_{\rm L}$ and polarization angle ψ. The marginalized likelihood-ratio is given by

$$\mathcal{L}^{\rm marg} = \frac{3}{\pi(D_{\rm max}^3 - D_{\rm min}^3)} \int_{D_{\rm min}}^{D_{\rm max}} D_{\rm L}^2 dD_{\rm L} \int_0^\pi d\psi \mathcal{L}. \tag{1}$$

1 Likelihood ratio

The logarithm of likelihood is given by

$$\ln \mathcal{L} = \sum_k \left[(d_k, h_k)_k - \frac{1}{2}(h_k, h_k)_k \right], \tag{2}$$

where k indexes each detector, d_k represents data from the k-th detector, and h_k represents a signal model at the k-th detector. The inner product is defined as

$$(x, y)_k \equiv 4\Re \left[\int_{f_{\rm low}}^{f_{\rm high}} df \frac{\tilde{x}^*(f)\tilde{y}(f)}{S_k(f)} \right], \tag{3}$$

where $f_{\rm low}$ and $f_{\rm high}$ are the low-frequency and high-frequency cutoffs of the analysis, and $S_k(f)$ is the one-sided power spectral density of the k-th detector.

The signal model is given by

$$\tilde{h}_k(f) = \frac{D_{\rm ref}}{D_{\rm L}} \left(F_{+,k}\tilde{h}_+(f) + F_{\times,k}\tilde{h}_\times(f) \right) {\rm e}^{-2\pi {\rm i} f t_k}, \tag{4}$$

where $D_{\rm ref}$ is a reference luminosity distance, $F_{+,k}$ and $F_{\times,k}$ are the beam pattern functions of the k-th detector, and t_k is the arrival time at the k-th detector. The beam pattern functions depend on ψ as follows,

$$F_{+,k} = \bar{F}_{+,k}\cos 2\psi + \bar{F}_{\times,k}\sin 2\psi, \quad F_{\times,k} = -\bar{F}_{+,k}\sin 2\psi + \bar{F}_{\times,k}\cos 2\psi, \tag{5}$$

where $\bar{F}_{+,k}$ and $\bar{F}_{+,k}$ are the values of $F_{+,k}$ and $F_{\times,k}$ at $\psi = 0$ respectively. Their squares and product are given by

$$\begin{aligned} (F_{+,k})^2 &= s_k + d_k \cos 4\psi + p_k \sin 4\psi, \\ (F_{\times,k})^2 &= s_k - d_k \cos 4\psi - p_k \sin 4\psi, \\ F_{+,k}F_{\times,k} &= p_k \cos 4\psi - d_k \sin 4\psi, \end{aligned} \tag{6}$$

where

$$s_k \equiv \frac{1}{2}\left[(\bar{F}_{+,k})^2 + (\bar{F}_{\times,k})^2 \right], \quad d_k \equiv \frac{1}{2}\left[(\bar{F}_{+,k})^2 - (\bar{F}_{\times,k})^2 \right], \quad p_k \equiv \bar{F}_{+,k}\bar{F}_{\times,k}. \tag{7}$$

Substituting (**??**) into the linear inner product $(d_k, h_k)_k$, we obtain

$$(d_k, h_k)_k = \frac{D_{\rm ref}}{D_{\rm L}} \left(A_k \cos 2\psi + B_k \sin 2\psi \right), \tag{8}$$

where

$$A_k \equiv \bar{F}_{+,k} L_{+,k} + \bar{F}_{\times,k} L_{\times,k}, \quad B_k \equiv \bar{F}_{\times,k} L_{+,k} - \bar{F}_{+,k} L_{\times,k}, \tag{9}$$

and

$$\begin{aligned} L_{+,k} &\equiv 4\Re \left[\int_{f_{\rm low}}^{f_{\rm high}} df \frac{\tilde{d}_k^*(f) \tilde{h}_+(f)}{S_k(f)} \mathrm{e}^{-2\pi \mathrm{i} f t_k} \right], \\ L_{\times,k} &\equiv 4\Re \left[\int_{f_{\rm low}}^{f_{\rm high}} df \frac{\tilde{d}_k^*(f) \tilde{h}_\times(f)}{S_k(f)} \mathrm{e}^{-2\pi \mathrm{i} f t_k} \right]. \end{aligned} \tag{10}$$

Substituting (**??**) into the quadratic inner product $(h_k, h_k)_k$, we obtain

$$(h_k, h_k)_k = \left(\frac{D_{\rm ref}}{D_{\rm L}} \right)^2 \left(C_k \cos 4\psi + D_k \sin 4\psi + E_k \right), \tag{11}$$

where

$$\begin{aligned} C_k &\equiv d_k (Q_{++,k} - Q_{\times\times,k}) + 2 p_k Q_{+\times,k}, \\ D_k &\equiv p_k (Q_{++,k} - Q_{\times\times,k}) - 2 d_k Q_{+\times,k}, \\ E_k &\equiv s_k (Q_{++,k} + Q_{\times\times,k}), \end{aligned} \tag{12}$$

and

$$Q_{++,k} \equiv (h_+, h_+)_k, \quad Q_{\times\times,k} \equiv (h_\times, h_\times)_k, \quad Q_{+\times,k} \equiv (h_+, h_\times)_k. \tag{13}$$

We have the following inequality for C_k, D_k and E_k,

$$E_k^2 - C_k^2 - D_k^2 = 4 s_k^2 (Q_{++,k} Q_{\times\times,k} - Q_{+\times,k}^2) \geq 0, \tag{14}$$

where the last inequality follows from the Schwartz inequality. The equality is established iff

$$\bar{F}_{+,k} = \bar{F}_{\times,k} = 0, \tag{15}$$

or there exist $\tilde{h}(f)$ and $0 \leq \phi \leq \pi$ such that

$$\tilde{h}_+(f) = \cos\phi \tilde{h}(f), \quad \tilde{h}_\times(f) = \sin\phi \tilde{h}(f). \tag{16}$$

Summing them up over detectors, we obtain

$$\ln \mathcal{L} = \frac{D_{\rm ref}}{D_{\rm L}} \left(A \cos 2\psi + B \sin 2\psi \right) - \frac{1}{2} \left(\frac{D_{\rm ref}}{D_{\rm L}} \right)^2 \left(C \cos 4\psi + D \sin 4\psi + E \right), \tag{17}$$

where

$$A = \sum_k A_k, \quad B \equiv \sum_k B_k, \quad C \equiv \sum_k C_k, \quad D \equiv \sum_k D_k, \quad E \equiv \sum_k E_k. \tag{18}$$

Eq. (**??**) leads to the following inequality,

$$E^2 - C^2 - D^2 = \sum_k (E_k^2 - C_k^2 - D_k^2) + 2\sum_{k>l}(E_k E_l - C_k C_l - D_k D_l) \tag{19}$$

$$\geq 2\sum_{k>l}\left(\sqrt{C_k^2 + D_k^2}\sqrt{C_l^2 + D_l^2} - C_k C_l - D_k D_l\right). \tag{20}$$

We also have the following inequality,

$$(C_k^2 + D_k^2)(C_l^2 + D_l^2) - (C_k C_l + D_k D_l)^2 = (C_k D_l - C_l D_k)^2 \geq 0. \tag{21}$$

Thus, the following inequality is established,

$$E^2 - C^2 - D^2 \geq 0, \tag{22}$$

where the equality is established iff

$$C_k D_l - C_l D_k = 0, \quad C_k C_l + D_k D_l \geq 0, \tag{23}$$

are satisfied for each pair (k, l) in addition to (**??**) or (**??**).

Let us investigate in which case the equality of (**??**) is established in more detail. If (**??**) is satisfied for each detector, (**??**) is satisfied, and the equality is trivially established. If (**??**) is satisfied, the quadratic inner products are given as follows,

$$Q_{++,k} = \cos^2\phi Q_k, \quad Q_{+\times,k} = \cos\phi\sin\phi Q_k, \quad Q_{\times\times,k} = \sin^2\phi Q_k, \tag{24}$$

where $Q_k \equiv (h, h)_k$. Thus, C_k and D_k are given by

$$C_k = (\cos 2\phi d_k + \sin 2\phi p_k)Q_k, \quad D_k = (\cos 2\phi p_k - \sin 2\phi d_k)Q_k. \tag{25}$$

Thus, the first condition of (**??**) leads to

$$C_k D_l - C_l D_k = (d_k p_l - d_l p_k)Q_k Q_l = 0. \tag{26}$$

We do not consider the case where gravitational waves are vanishing and assume $Q_k > 0$. Then, we obtain

$$d_k p_l - d_l p_k = \frac{1}{2}(\bar{F}_{+,k}\bar{F}_{+,l} + \bar{F}_{\times,k}\bar{F}_{\times,l})(\bar{F}_{+,k}\bar{F}_{\times,l} - \bar{F}_{\times,k}\bar{F}_{+,l}) = 0. \tag{27}$$

The second condition of (**??**) leads to

$$C_k C_l + D_k D_l = (d_k d_l + p_k p_l)Q_k Q_l \geq 0, \tag{28}$$

which leads to

$$d_k d_l + p_k p_l = \frac{1}{4}\left(\bar{F}_{+,k}\bar{F}_{+,l} + \bar{F}_{\times,k}\bar{F}_{\times,l}\right)^2 - \frac{1}{4}\left(\bar{F}_{+,k}\bar{F}_{\times,l} - \bar{F}_{\times,k}\bar{F}_{+,l}\right)^2 \geq 0. \tag{29}$$

Eq. (**??**) and (**??**) lead to

$$\bar{F}_{+,k}\bar{F}_{\times,l} - \bar{F}_{\times,k}\bar{F}_{+,l} = 0. \tag{30}$$

Substituting the following representations of beam pattern functions into this equation,

$$\bar{F}_{+,k} = \bar{F}_k \cos\theta_k, \quad \bar{F}_{\times,k} = \bar{F}_k \sin\theta_k, \tag{31}$$

we obtain

$$\theta_l = \theta_k + n\pi. \ \ (n \in \mathbb{Z}) \tag{32}$$

Since it is satisfied for each pair (k, l), there exists a common phase θ such that

$$\bar{F}_{+,k} = (-1)^{s_k}\bar{F}_k \cos\theta, \quad \bar{F}_{\times,k} = (-1)^{s_k}\bar{F}_k \sin\theta, \tag{33}$$

where $s_k = 0$ or 1. Thus, in the case where (**??**) is not satisfied, the equality of (**??**) is established iff (**??**) and (**??**) are satisfied.

2 Coordinate transformation

To simply the likelihood, we express it in the following x–y coordinate,

$$x \equiv \frac{D_{\rm ref}}{D_{\rm L}} \cos 2\psi, \qquad y \equiv \frac{D_{\rm ref}}{D_{\rm L}} \sin 2\psi. \tag{34}$$

In addition, we use the following parametrization of (A, B) and (C, D),

$$A = R\cos\zeta, \quad B = R\sin\zeta, \quad (R > 0,\ 0 \le \zeta < 2\pi) \tag{35}$$

$$C = S\cos 2\xi, \quad D = S\sin 2\xi. \quad (S > 0,\ 0 \le \xi < \pi) \tag{36}$$

With the coordinate transformation, likelihood is simplified to Gaussian distribution,

$$\ln\mathcal{L} = -\frac{1}{2}\begin{pmatrix} x & y \end{pmatrix}\Gamma\begin{pmatrix} x \\ y \end{pmatrix} + R\begin{pmatrix} \cos\zeta & \sin\zeta \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}, \tag{37}$$

where

$$\Gamma = \begin{pmatrix} E + S\cos 2\xi & S\sin 2\xi \\ S\sin 2\xi & E - S\cos 2\xi \end{pmatrix}. \tag{38}$$

Γ is diagonalized as follows,

$$\Gamma = U^T\Lambda U, \quad U = \begin{pmatrix} \cos\xi & \sin\xi \\ -\sin\xi & \cos\xi \end{pmatrix}, \quad \Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \tag{39}$$

where

$$\lambda_1 = E + S, \quad \lambda_2 = E - S. \tag{40}$$

Thus, with the following coordinate transformation,

$$\begin{pmatrix} s \\ t \end{pmatrix} \equiv U \begin{pmatrix} x \\ y \end{pmatrix}, \tag{41}$$

$\ln \mathcal{L}$ can be simplified as follows,

$$\ln \mathcal{L} = -\frac{\lambda_1}{2} s^2 - \frac{\lambda_2}{2} t^2 + R\cos(\zeta - \xi) s + R\sin(\zeta - \xi) t. \tag{42}$$

The coefficients of the linear terms are calculated as

$$U \begin{pmatrix} \cos\zeta \\ \sin\zeta \end{pmatrix} = \begin{pmatrix} \cos(\zeta - \xi) \\ \sin(\zeta - \xi) \end{pmatrix}. \tag{43}$$

Eq. (**??**) leads to

$$\lambda_1 \geq \lambda_2 \geq 0. \tag{44}$$

If $\lambda_2 > 0$, the squares can be completed as follows,

$$\ln \mathcal{L} = -\frac{\lambda_1}{2}(s - \bar{s})^2 - \frac{\lambda_2}{2}(t - \bar{t})^2 + \frac{\lambda_1}{2}\bar{s}^2 + \frac{\lambda_2}{2}\bar{t}^2, \tag{45}$$

where

$$\bar{s} = \frac{R\cos(\zeta - \xi)}{\lambda_1}, \qquad \bar{t} = \frac{R\sin(\zeta - \xi)}{\lambda_2}. \tag{46}$$

Next, we consider the case of $\lambda_1 > 0,\ \lambda_2 = 0$. λ_2 vanishes iff (**??**) is satisfied, or (**??**) and (**??**) are satisfied. Since λ_1 also vanishes for the first case, the second conditions need to be satisfied. Eq (**??**) and (**??**) lead to

$$\frac{A}{\sqrt{A^2 + B^2}} = \cos(\theta - \phi), \quad \frac{B}{\sqrt{A^2 + B^2}} = \sin(\theta - \phi), \tag{47}$$

$$\frac{C}{\sqrt{C^2 + D^2}} = \cos 2(\theta - \phi), \quad \frac{D}{\sqrt{C^2 + D^2}} = \sin 2(\theta - \phi). \tag{48}$$

Thus, we have

$$\zeta = \xi = \theta - \phi, \tag{49}$$

and the linear term of t in $\ln L$ vanishes. After the square is completed, $\ln L$ is given by

$$\ln \mathcal{L} = -\frac{\lambda_1}{2}(s - \bar{s})^2 + \frac{\lambda_1}{2}\bar{s}^2. \tag{50}$$

Finally, we consider the case of $\lambda_1 = \lambda_2 = 0$. This leads to $E = 0$, which is satisfied iff (**??**) is satisfied for each detector. Thus, all of A, B, C and D vanish, and $\ln \mathcal{L}$ vanishes,

$$\ln \mathcal{L} = 0. \tag{51}$$

In summary, $\ln \mathcal{L}$ is given by

$$\ln \mathcal{L} = \begin{cases} -\frac{\lambda_1}{2}(s - \bar{s})^2 - \frac{\lambda_2}{2}(t - \bar{t})^2 + \frac{\lambda_1}{2}\bar{s}^2 + \frac{\lambda_2}{2}\bar{t}^2, & (\lambda_1 \geq \lambda_2 > 0) \\ -\frac{\lambda_1}{2}(s - \bar{s})^2 + \frac{\lambda_1}{2}\bar{s}^2, & (\lambda_1 > \lambda_2 = 0) \\ 0. & (\lambda_1 = \lambda_2 = 0) \end{cases} \tag{52}$$

3 Marginalization

Finally, we calculate $\mathcal{L}^{\rm marg}$. Since the likelihood for $\lambda_1 = \lambda_2 = 0$ is trivial, we focus on the case of $\lambda_1 > 0$. Considering $\bar{t}$ as a free parameter which does not depend on λ_2, we can treat the case of $\lambda_1 > \lambda_2 = 0$ as a limit of $\lambda_1 > \lambda_2 > 0$ with $\lambda_2 \to +0$. Thus, we consider the following form of $\ln \mathcal{L}$,

$$\ln \mathcal{L} = -\frac{\lambda_1}{2}(s-\bar{s})^2 - \frac{\lambda_2}{2}(t-\bar{t})^2 + \frac{\lambda_1}{2}\bar{s}^2 + \frac{\lambda_2}{2}\bar{t}^2. \quad (\lambda_1 \geq \lambda_2 \geq 0,\ \lambda_1 > 0) \tag{53}$$

Substituting this into (**??**), we obtain

$$\mathcal{L}^{\rm marg} = \frac{3D_{\rm ref}^3}{2\pi(D_{\rm max}^3 - D_{\rm min}^3)} \mathrm{e}^{\frac{\lambda_1}{2}\bar{s}^2 + \frac{\lambda_2}{2}\bar{t}^2} I, \tag{54}$$

where

$$I \equiv \int_{D_{\rm ref}/D_{\rm max} < \sqrt{(s+\bar{s})^2+(t+\bar{t})^2} < D_{\rm ref}/D_{\rm min}} \frac{\mathrm{e}^{-\frac{\lambda_1}{2}s^2 - \frac{\lambda_2}{2}t^2}}{\left[(s+\bar{s})^2 + (t+\bar{t})^2\right]^{\frac{5}{2}}} dsdt. \tag{55}$$

Here we have performed the translation: $s \to s + \bar{s},\ t \to t + \bar{t}$.

Typically, $D_{\rm min}$ is small enough to encompass the true value of $D_{\rm L}$. It means likelihood is exponentially suppressed at $D_{\rm L} < D_{\rm min}$. Thus, extending the upper limit of integral to infinity $D_{\rm ref}/D_{\rm min} \to \infty$ gives negligible differences in the results. On the other hand, the integral is divergent in the limit of $D_{\rm ref}/D_{\rm max} \to 0$. This reflects the fact that likelihood-ratio converges to unity while we get infinite prior volume at $D_{\rm L} > D_{\rm max}$. To prevent from getting the infinite prior volume, we add a regulator term $k(D_{\rm ref}/D_{\rm max})^2$ in the denominator, where $k > 0,\ k = \mathcal{O}(1)$. Then, we obtain the following approximate form,

$$I \simeq \int_{-\infty}^{\infty} \frac{\mathrm{e}^{-\frac{\lambda_1}{2}s^2 - \frac{\lambda_2}{2}t^2}}{\left[(s+\bar{s})^2 + (t+\bar{t})^2 + k(D_{\rm ref}/D_{\rm max})^2\right]^{\frac{5}{2}}} dsdt. \tag{56}$$

(What value should be used as k? Do we need to subtract something compensating for the extension?)